Method of automatic identification of ruler scales on images to solve the problem of estimating the geometric dimensions of certain objects

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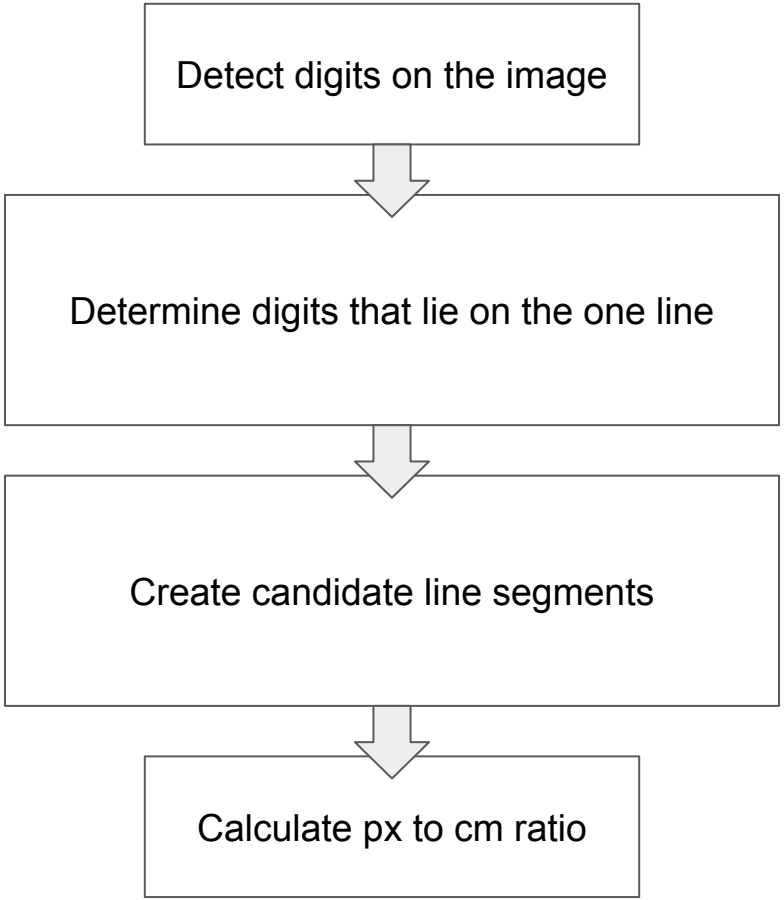


https://arxiv.org/pdf/2201.00414.pdf





https://www.researchgate.net/publication/325422545_Implementation_features_of_wounds_visual_comparison_subsystem


Detect digits on the image
Determine digits that lie on the one line
Create candidate line segments
Calculate px to cm ratio

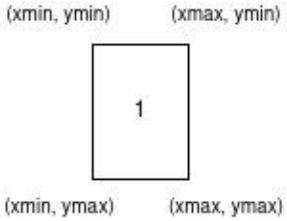
Detect digits on the image

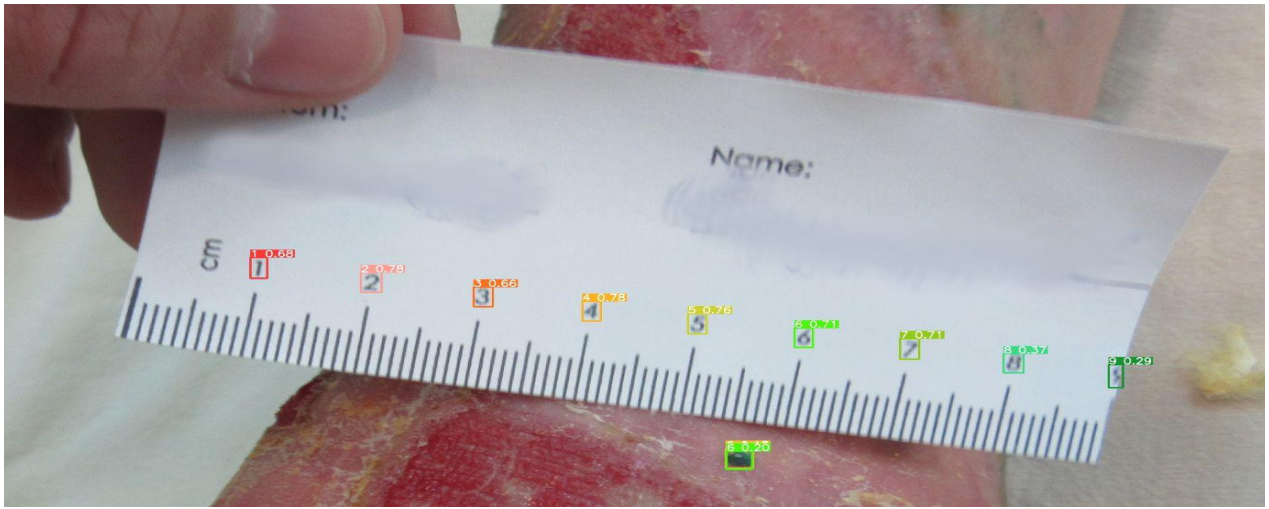
Determine digits that lie on the one line

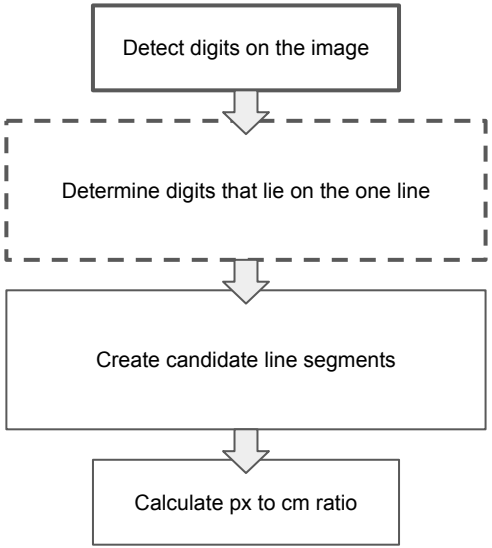
Create candidate line segments

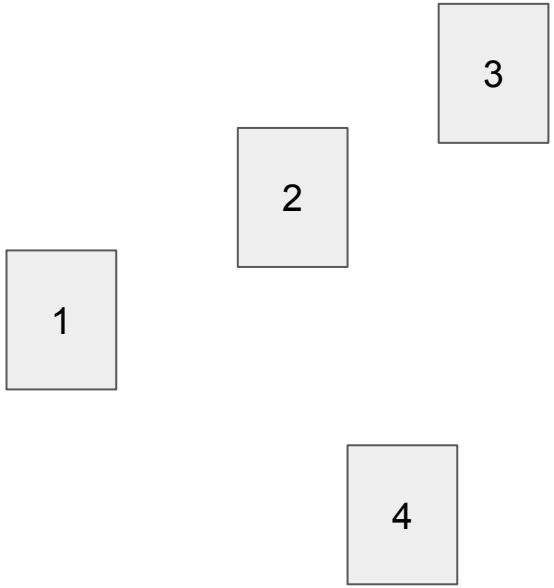
Calculate px to cm ratio

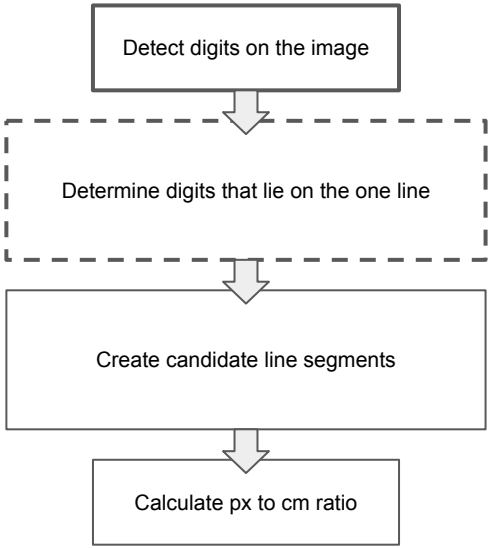
any digit detection method that can produce output in the desired format:

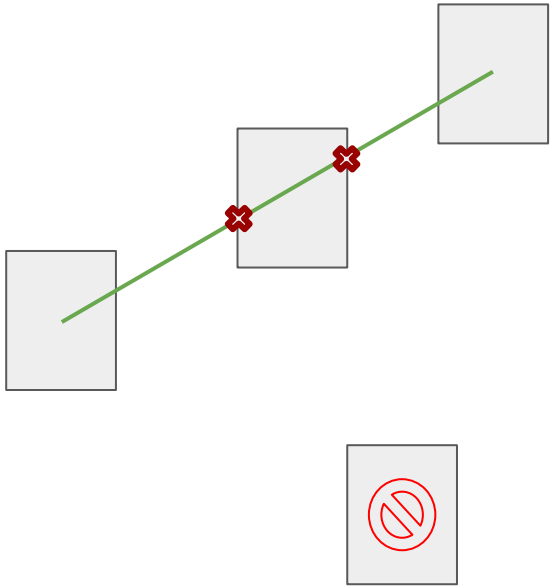


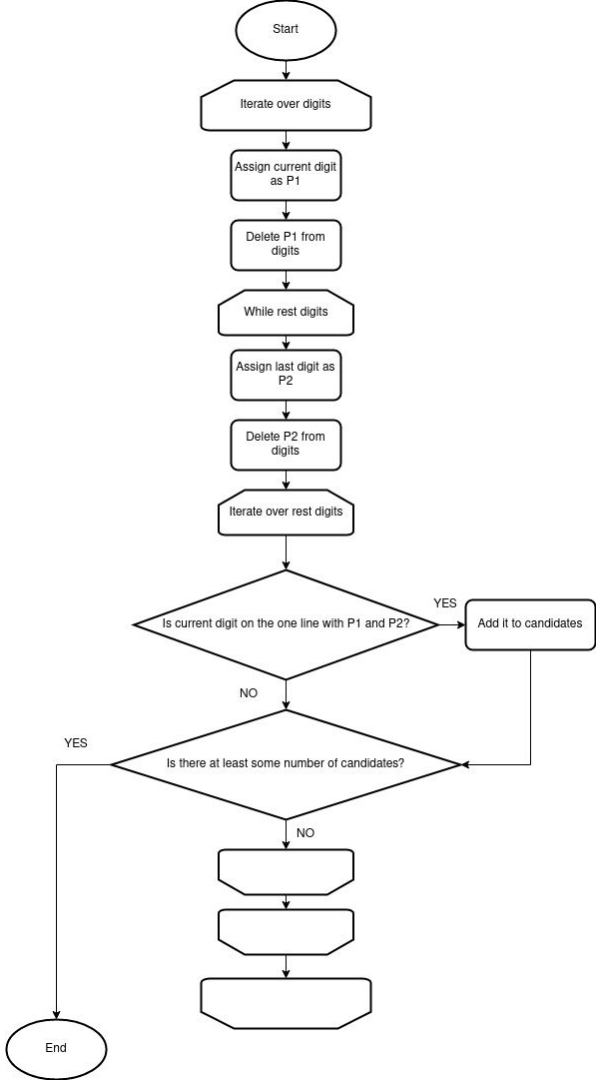



Detect digits on the image
Determine digits that lie on the one line
Create candidate line segments
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3
2
1
4


Detect digits on the image
Determine digits that lie on the one line
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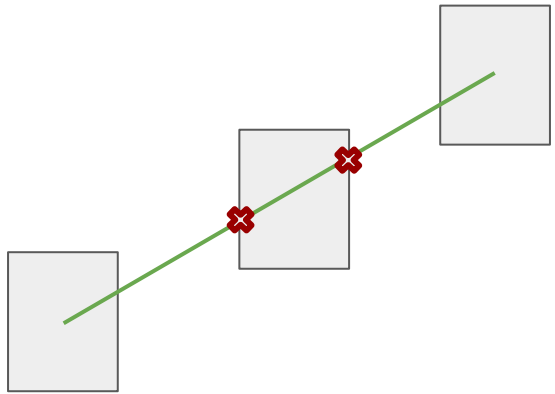

Start
Iterate over digits
Assign current digit as P1
Delete P1 from digits
While rest digits
Assign last digit as P2
Delete P2 from digits
Iterate over rest digits
Is current digit on the one line with P1 and P2?
YES
Add it to candidates
NO
YES
Is there at least some number of candidates?
NO
End

Detect digits on the image

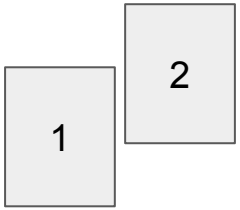
Determine digits that lie on the one line

Create candidate line segments

Calculate px to cm ratio



?

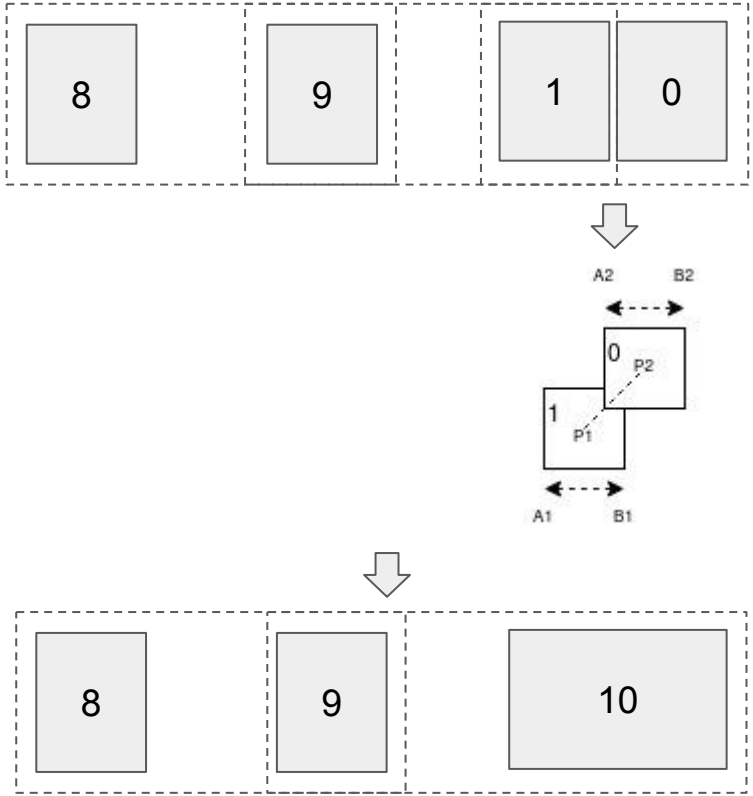


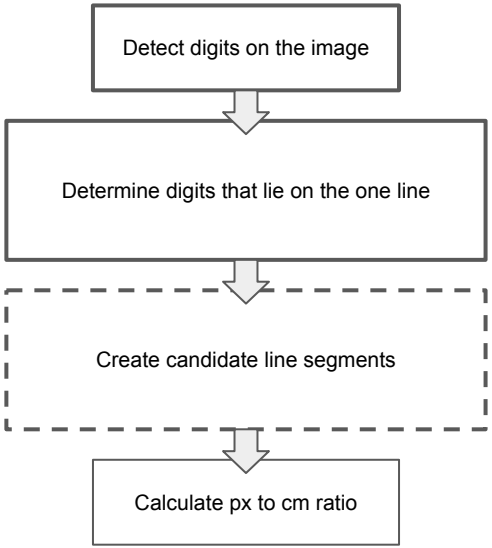
Detect digits on the image

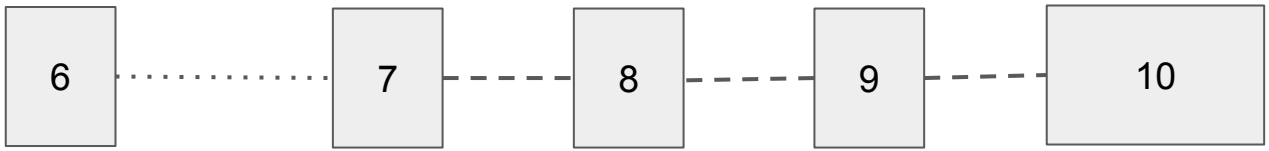
Determine digits that lie on the one line

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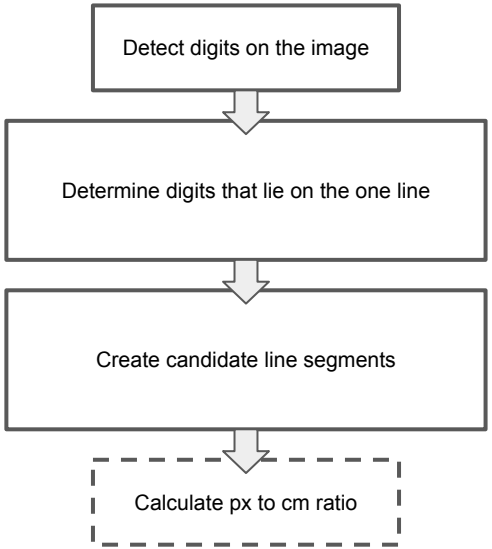




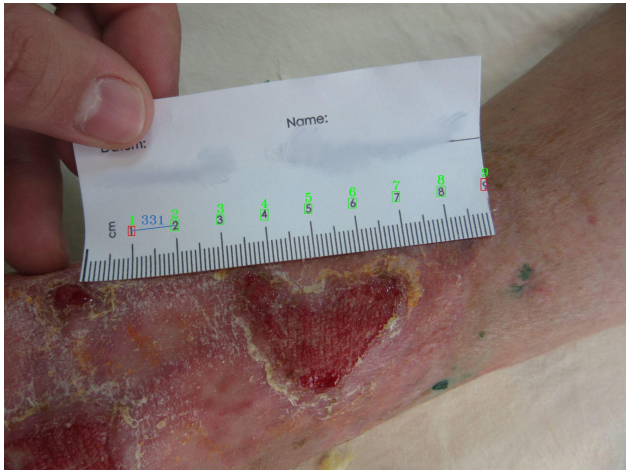
segments of one length

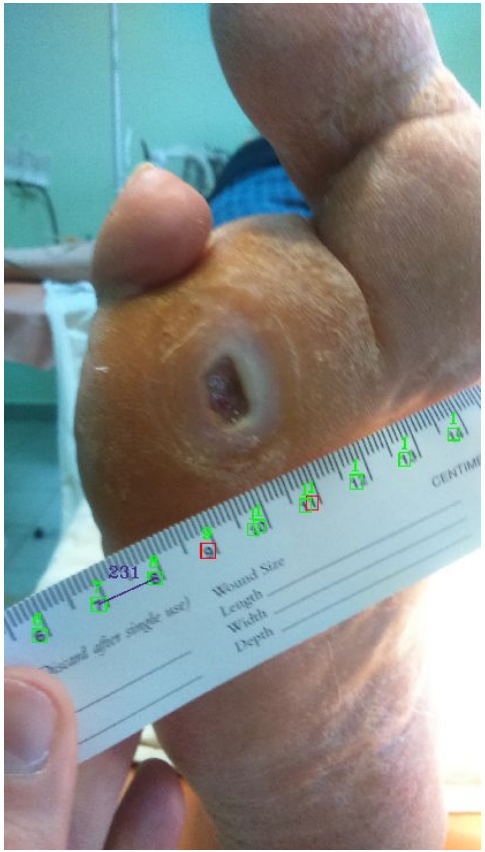
segments of the other length

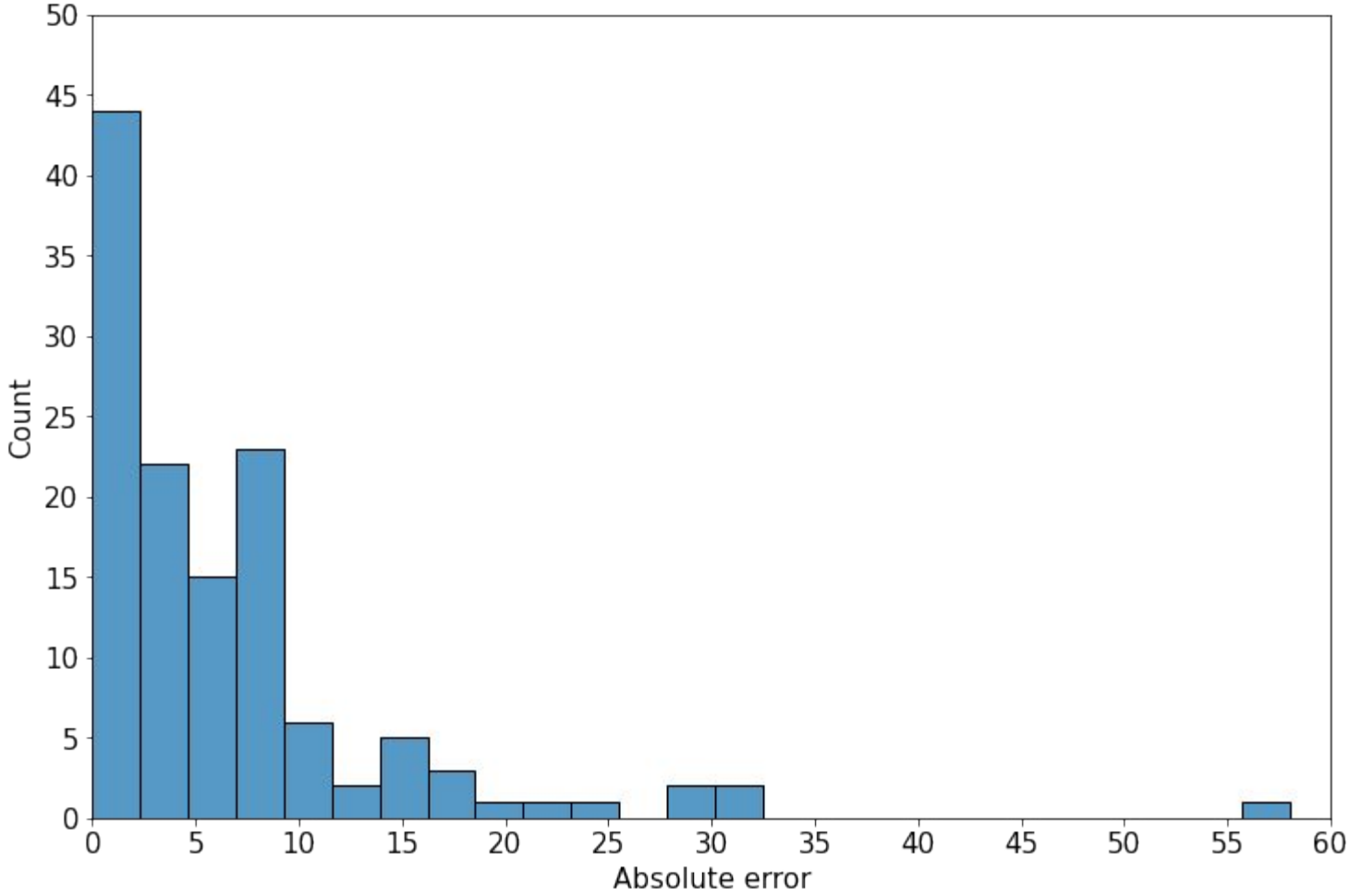
and there may be many more segment groups, but we take the largest



use the distance between centers of two consecutive digits from the largest segment group as pixel to cm ratio...







(with hand measurement spread of the results for one sample may be in range of even 20-30 pixels)

- sensitivity to the method of digit detection
- lots of iterations)
- hard to determine the number of digits (that lie on the same line) to search for
- intuitive approach
- error comparable and even outperforms hand measurement

Thank you for the attention.



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